Backtests

How qp replays a strategy over many days on the stocks the screener tools actually found, the honesty rules it keeps, every setting of a run, the accounting it can apply, and what it reports. Written from `quant-platform/chart/backtest.py`, `report.py`, `rank_sweep.py`, `store.py` and the Backtest section of `chart/static/index.html`.

Related: the strategy grammar and fill models (STRATEGIES.md), the bar feeds (DATA-FEEDS.md), the CANSLIM letters as of a day (CANSLIM.md).

What a run does

For every trading day in the range, and every symbol in that day's universe, it calls **the same** `strategy.evaluate()` **the chart preview uses**, as of that day. There is no separate backtest evaluator, so the preview and the backtest cannot disagree.

- **One run at a time** (the box is small). A second start while one runs is refused with the number of the running one. A run can be stopped; it stops at its next symbol. Closing the page does not stop a run: reopening it picks the run up again. If qp itself restarts, the run is marked interrupted.
- Sequential: one (day, symbol) pair at a time, progress reported as the share of pairs done.
- **Every run stores a frozen copy of the strategies it ran**, so editing a strategy later never changes what an old result means.

The universe

`universe.kind`	**Each day evaluates**
`symbols`	the same fixed list every day
`register`	one register (R1 or Shortlist) **as that tool froze it that day**
`tools`	the registers of every tool assigned to the strategy (or named in the spec), unioned; with two books (a long and a short strategy) their tools are unioned too

`extra_symbols` adds names to a register universe; their trades are marked `_extra`, so "not on the register" stays visible.

Honesty rules (all on by default)

Rule	What it prevents
As of the day. Each (day, symbol) is evaluated with `asof` = that day; only trades **entered on that day** count. Warm-up days are context, never trades	look-ahead; double-counting warm-up signals
The watchlist gate (`scan_gate`): a trade the strategy took **before the scanner had found the stock** is dropped. `scan_gate: false` shows the old numbers on purpose. No effect on a `symbols` universe	trades no alert could have fired
The desk fill (`fill: desk`): next bar's open, levels from the decision close (STRATEGIES.md)	the optimistic `close` fill (the old default)
Costs: `cost_bps` per side (spread + slippage + commission; a round trip pays twice), and per-share fees with a per-order minimum	"a backtest without costs lies"
Open at the end: a position still open when its window ends is recorded as `open` with its unrealised return, **never mixed into the win rate**	flattering win rates
Coverage, stated: pairs evaluated, pairs whose feed returned no bars, pairs per scanning tool, how many each filter removed	a thin feed or one dominant tool passing for a strategy that rarely fires

The settings of a run

What to test

Spec	Meaning	Default
`strategy_id` / `strategy` / `strategies`	one strategy, or two books (long and short)	—
`universe`	above	register R1
`start`, `end`	the date range	—
`tf`	bar timeframe (1m, 2m, 5m, 15m, 30m, 1h, 1d)	5m
`feed`	bar feed (DATA-FEEDS.md)	polygon
`view`	`all` (extended hours) or `regular`	all
`fill`	`desk`, `next_open`, `close`	**desk**
`days`	warm-up sessions before each day (auto-extended)	3
`rules`	session rules: regular-hours entries, end-of-day close, entries per day	none

Filters and caps (opt-in)

Spec	Does
`min_rvol`	In-Play filter: a pair is evaluated only if its relative volume, **measured by qp at the strategy's window start**, is above this (not the register's one-bar snapshot); below and unverifiable are counted apart
`canslim`	a gate on the CANSLIM letters **as known on that day** (`chart/canslim.py`); a rule naming a letter that cannot be rebuilt for a past day is refused before the run starts; blocked and unverifiable pairs are counted. **API only: the page has no control for it** (Q-N4)
`check_shortable`	drops shorts the broker would not lend (Alpaca). Alpaca keeps no borrow history, so **today's** flag is applied to a past day, and the report says so; a name it cannot answer for is kept and named
`rank_per_day` + metric	keep the best N signals a day. Metrics: `vwap_extension` (distance from decision price to stop, largest first: right only for a VWAP-anchored stop, the T2 setup), `tight_stop` (the same, smallest first), `reg_score` (the register's score), `rvol` (qp's RVOL at the window start). **No default: an unnamed or unknown metric is an error**, because the wrong one silently reshapes the strategy (backtest #231 lost 103 of 117 signals to it)
desk caps	**one entry per stock per day**, and a whole-run trades-per-day cap, applied in the order signals fired

`chart/rank_sweep.py` runs the same selection over a finished run's trades with every metric, to measure which one suits a strategy.

Money: choose one accounting

Block	Inputs	Answers
unit	none	per-trade % returns
account	`account_equity` (default 100,000 on the page), `risk_pct` (0.5) **or** `risk_usd` (not both), `max_position_pct`, `max_leverage` (1 = cash), `size_ratio` (this account's share of the standard size), `max_order_value`, `min_shares`	real money: shares from the stop so every stop-out loses the same %; equity **compounds** in trade order; trades with no usable stop are **excluded and counted**, never sized at 1 share; trades the buying power could not take are skipped and counted
TTP (prop firm, Trade The Pool defaults)	`fee_per_share`, `fee_min` per order, `shares` (fixed size, default 100), `min_profit_ps`	dollar P&L for a fixed size; the **min-profit rule**: a winner under `min_profit_ps` $/share does not count towards the target (losses always count), tested on the position's average across all its fills; every fill pays its own order fee
challenge (on top of an account run)	`challenge: {target_pct, max_dd_pct, basis: start | peak}`	did the account reach the target **before** it ever touched the drawdown limit? Also walks a worst case in which every open position stops out at once, because a closed-trade curve cannot see a position that dipped and came back

What a run reports

Route	Gives
`GET /api/backtest/{id}`	status, progress, summary, trades
`GET /api/backtests`	every run, newest first (`brief=1` for headlines only)
`GET /api/backtest/{id}/report`	a self-contained, phone-readable HTML report
`GET /api/backtest/{id}/stats.json`	every number in the report
`GET /api/backtest/{id}/csv`	every trade, with every register column of its card
`GET /api/backtest/{id}/journal.csv`	one row per trade for a spreadsheet or the journal
`POST /api/backtest/{id}/stop` , `DELETE /api/backtest/{id}`	stop, delete

On the page, each trade in the list opens that symbol and day on the chart with the run's own timeframe, feed, fill model and strategy. A filter (`=` , `≠` , `>` , `<` , `in` , `has` ; times as `hhmm` , `dur_min` , `dow`) narrows the trades and the filtered CSV.

The report (`report.py`)

Computed afterwards from the stored trades, so it can be re-run on any past backtest. It says whether every number is in **dollars** (an account run) or **unit %**.

Section	Numbers
headline	trades, wins, losses, breakeven (exactly 0 is not a win), win rate, gross profit and loss, net profit, **profit factor** (none with no losing trade, never ∞), expectancy, average win and loss, payoff ratio, largest win and loss
R-multiples	total, average, best, worst, and the distribution in buckets. **R reads from net**, the same money as net profit; the gross total is shown beside it
holding time	average, longest, shortest, winners vs losers, total time in positions
splits	by side, by exit reason, by symbol, by day
equity and risk	the curve marked at **exit**, maximum drawdown in $ and %, its length, streaks
ratios	Sharpe, Sortino, Calmar from daily returns over **every session in the range** (flat days included), annualised and flagged as extrapolation on short samples
exposure	share of regular-hours minutes in a position
sample adequacy	each warning as a fact and its reason
the borrow check	states when “check shortable” was ticked but nothing was answered (backtest #355)
prop-firm detail	counted vs wasted profit under the min-profit rule

Found in this review

#	What
Q-N4	The CANSLIM gate exists in the engine but has no control on the page; it can only be set through the API